

Elouan Charbonnier

Threat Analyst — Malware Analysis & Reverse Engineering

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🌐 halftimeoflife.github.io | 🚩 RootMe — 2740 pts | 🚗 Driving licence — French national

Profile

Master's graduate in Cryptology and Computer Security, specialising in malware analysis and reverse engineering. My internship at eShard on anti-emulation techniques and my published projects allowed me to develop an approach covering static and dynamic analysis, evasion technique detection, and the production of actionable threat intelligence. I am targeting roles as a malware analyst, reverse engineer, or CTI analyst.

Education

University of Bordeaux

MSc Cryptology & Computer Security

Bordeaux

2023–2025

University of Bordeaux

BSc Mathematics & Computer Science

Bordeaux

2020–2023

Certification

Centri

BTL1 — Blue Team Level 1 (Security Blue Team)

expected July 2026

Professional Experience

🚀 eShard

Pessac, France

Implementation, Detection and Bypass of Anti-Emulation Techniques on Windows

Mar–Aug 2025

Reverse engineering internship focused on analysing, implementing and bypassing anti-VM/anti-emulation techniques within eShard's TTD-based platform (esReverse).

- Reverse-engineered malware samples to identify anti-emulation markers (Ghidra, Binary Ninja)
- Bypassed anti-VM protections via Windows API patching (WinDbg) and QEMU configuration tuning
- Implemented a wide range of anti-VM techniques (CPUID, RDTSC, Windows API, ...) in x86 assembly and C
- Designed an automated binary generation pipeline embedding anti-VM techniques, used to validate detection coverage
- Automated detection of suspicious behaviours through TTD scripting (Python)
- Produced a technical article published on eShard's blog and a YouTube playlist
- Documented all research under Jupyter Notebooks

Projects

🔗 GhidraMAT: Ghidra scripting framework for automated static detection of malicious TTPs. Speeds up sample triage by producing directly usable JSON/TXT reports.

🔗 Malauto: WinDbg extension automating dynamic analysis of packer behaviours on Windows.

🔗 Wine APT29: Full analysis of WINELOADER/GRAPELOADER (APT29, 2025): execution flow reconstruction, API deobfuscation, C2 identification, and evasion technique mapping.

🔗 Panoptiv: Windows API hooking DLL designed to bypass anti-VM protections.

Publications

2025: Thesis — “Detection of Anti-VM Protections in Malware” [PDF]

2025: Article — “Anti-Emulation Techniques on Windows” — eShard [Article]

2025: Report — “Linux Kernel Rootkits in 2025” [PDF]

Technical Skills

OS: Linux, Windows

Languages: Python, C, x86/x64 Assembly, $\LaTeX$

Virtualisation: QEMU, VMware, VirtualBox, FLARE-VM

Static analysis: Ghidra, Binary Ninja

Dynamic analysis: WinDbg, GDB, Reven TTD

CTI: MITRE ATT&CK, YARA

Languages

French: Native

English: B2 — Linguaskill certification

Interests

🧑‍🦴: Hiking

⚡: Bivouacking